

MSc in Engineering – Mechatronics and Automation

Term Thesis

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NDA (Non-Disclosure Agreement) yes
Type of work MTh
ECTS 30

Subject Project Title

Abstract

The Swiss watch industry requires flexible and efficient automation solutions to remain competitive. Mobile robots and mobile manipulators (MoMa) have the potential to automate internal logistics and material handling while improving process efficiency and traceability. However, their integration into existing production environments remains challenging due to high precision requirements, limited workspace, and the need for standardized communication.

This thesis aims to develop a manufacturer-independent integration concept for robotic systems within the automation projects of ETA SA.

Objectives

- Develop a standardized robot integration architecture.
- Create a common communication interface between SCADA, an orchestrator, and robotic systems.
- Support different robot types (AMR, MoMa, and stationary industrial robots).
- Validate the concept through a Proof of Concept (PoC).

Expected Deliverables

- Functional demonstrator with an AMR and a robotic manipulator.
- Standardized communication interfaces.
- Modular software framework for robot integration.
- Technical documentation and user manual.
- Recommendations for future industrial deployment.

The proposed solution will establish a reusable and scalable software architecture, reducing engineering effort, simplifying future robot integration, and increasing flexibility in selecting robotic platforms for ETA SA and the Swatch Group.

List of project objectives

Literature

- [1] MSE: TSM_AutMobRoS
 [2] Siegwart, R: "Intro to Autonomous Mobile Robots", 2nd ed., 2011.
 [3] OpenSource Fleet Manager: <https://github.com/open-rmf/rmf.git>

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Project proposal by email 27.07.2026

Project start 15.07.2026 (required)

Report delivery* 22.1.2027 (required according to the deadlines)

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